

EXPERIMENT -3

AIM

To create an Employee–Department–Project relational database, insert at least 30 employees across 5 departments and 8 projects, and demonstrate SQL queries using Selection, Projection, Aggregate Functions, GROUP BY, HAVING, CASE expressions, and ORDER BY.

SOURCE CODE

```
CREATE DATABASE CompanyDB;

USE CompanyDB;

CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL
);

CREATE TABLE Project (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(100) NOT NULL,
    budget DECIMAL(12, 2),
    dept_id INT,
    FOREIGN KEY (dept_id)
        REFERENCES Department(dept_id)
);

CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50) NOT NULL,
    salary DECIMAL(10, 2),
    hire_date DATE,
    dept_id INT,
    project_id INT,
    FOREIGN KEY (dept_id)
        REFERENCES Department(dept_id),
    FOREIGN KEY (project_id)
        REFERENCES Project(project_id)
);
```

Insert Departments

```
INSERT INTO Department VALUES
(1, 'IT'),
(2, 'HR'),
(3, 'Finance'),
(4, 'Marketing'),
(5, 'Operations');
```

Insert Projects

```
INSERT INTO Project VALUES
(101, 'Cloud Migration', 150000, 1),
(102, 'AI Analytics', 200000, 1),
(103, 'Recruitment Portal', 80000, 2),
(104, 'Financial Dashboard', 120000, 3), (105, 'Digital Campaign', 90000, 4),
(106, 'Supply Chain System', 180000, 5),
(107, 'Mobile Application', 140000, 1),
```

```
(108,'Employee Wellness',60000,2);
```

Insert 30 Employees

```
INSERT INTO Employee VALUES
(1,'Aarav',65000,'2022-01-10',1,101),
(2,'Vivaan',72000,'2021-03-15',1,102),
(3,'Aditya',58000,'2023-06-20',1,107),
(4,'Arjun',81000,'2020-08-12',1,101),
(5,'Kabir',69000,'2022-11-05',1,102),
(6,'Reyansh',62000,'2024-02-18',1,107),

(7,'Ananya',52000,'2022-04-11',2,103),
(8,'Diya',48000,'2023-01-22',2,108), (9,'Myra',61000,'2021-07-19',2,103),
(10,'Sara',55000,'2022-09-30',2,108),
(11,'Ishita',47000,'2024-03-14',2,103),
(12,'Meera',59000,'2020-12-01',2,108),
(13,'Rohan',75000,'2021-02-17',3,104),
(14,'Karan',68000,'2022-05-09',3,104),
(15,'Nikhil',83000,'2019-10-21',3,104),
(16,'Yash',62000,'2023-08-13',3,104),
(17,'Manav',71000,'2021-11-28',3,104),
(18,'Dev',56000,'2024-01-09',3,104),

(19,'Aanya',54000,'2022-02-25',4,105),
(20,'Kiara',63000,'2021-06-16',4,105),
(21,'Tanya',57000,'2023-04-10',4,105),
(22,'Riya',66000,'2020-09-07',4,105),
(23,'Avni',51000,'2024-02-05',4,105),
(24,'Navya',70000,'2022-12-19',4,105),
(25,'Samar',60000,'2021-01-12',5,106),
(26,'Dhruv',73000,'2020-04-23',5,106),
(27,'Atharv',67000,'2022-07-18',5,106),
(28,'Parth',59000,'2023-05-29',5,106),
(29,'Rudra',76000,'2019-08-31',5,106),
(30,'Veer',64000,'2024-01-20',5,106);
```

1. Selection

Display employees whose salary is greater than 70,000.

```
SELECT *
FROM Employee
WHERE salary > 70000;
```

OUTPUT

I/O

AI Agent

83 ms

STDIN

Input for the program (Optional)

Output:

emp_id	emp_name	salary	hire_date	dept_id	project_id
2	Vivaan	72000.00	2021-03-15	1	102
4	Arjun	81000.00	2020-08-12	1	101
13	Rohan	75000.00	2021-02-17	3	104
15	Nikhil	83000.00	2019-10-21	3	104
17	Manav	71000.00	2021-11-28	3	104
26	Dhruv	73000.00	2020-04-23	5	106
29	Rudra	76000.00	2019-08-31	5	106

2. Projection

Display only employee names and salaries.

```
SELECT emp_name, salary
FROM Employee;
```

OUTPUT

I/O	AI Agent	95 ms	✓
STDIN			
Input for the program (Optional)			
Output:			
+-----+-----+			
emp_name salary			
+-----+-----+			
Aarav 65000.00			
Vivaan 72000.00			
Aditya 58000.00			
Arjun 81000.00			
Kabir 69000.00			
Reyansh 62000.00			
Ananya 52000.00			
Diya 48000.00			
Myra 61000.00			
Sara 55000.00			
Ishita 47000.00			
Meera 59000.00			
Rohan 75000.00			
Karan 68000.00			

3. Aggregate Functions

Find total employees, average salary, maximum salary, minimum salary and total salary.

```
SELECT
    COUNT(*) AS total_employees,
    AVG(salary) AS average_salary,
    MAX(salary) AS highest_salary,
    MIN(salary) AS lowest_salary,
    SUM(salary) AS total_salary
FROM Employee;
```

I/O	AI Agent	86 ms	✓
STDIN			
Input for the program (Optional)			
Output:			
+-----+-----+-----+-----+-----+			
total_employees average_salary highest_salary lowest_salary total_salary			
+-----+-----+-----+-----+-----+			
30 63466.666667 83000.00 47000.00 1904000.00			
+-----+-----+-----+-----+-----+			

4. GROUP BY

Display the number of employees and average salary in each department.

```
SELECT
    d.dept_name,
    COUNT(e.emp_id) AS employee_count,
    AVG(e.salary) AS average_salary
FROM Department d
JOIN Employee e
ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name;
```

OUTPUT

I/O	AI Agent	105 ms	✓
STDIN			
Input for the program (Optional)			
Output:			
+-----+-----+-----+			
dept_name employee_count average_salary			
+-----+-----+-----+			
IT	6	67833.333333	
HR	6	53666.666667	
Finance	6	69166.666667	
Marketing	6	60166.666667	
Operations	6	66500.000000	
+-----+-----+-----+			

5. HAVING

Display departments whose average salary is greater than 65,000.

```
SELECT
    d.dept_name,
    AVG(e.salary) AS average_salary
FROM Department d
JOIN Employee e
ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name
HAVING AVG(e.salary) > 65000;
```

OUTPUT

Department Average Salary

IT 67833.33

Finance 69166.67

Operations 66500.00

6. CASE Expression

Classify employees according to their salary.

```
SELECT
    emp_name,      salary,
    CASE
        WHEN salary >= 75000 THEN 'High Salary'
        WHEN salary >= 60000 THEN 'Medium Salary'
        ELSE 'Low Salary'
    END AS salary_category
FROM Employee;
```

OUTPUT

I/O	AI Agent	94 ms																																																
STDIN																																																		
Input for the program (Optional)																																																		
Output:																																																		
<table><thead><tr><th>emp_name</th><th>salary</th><th>salary_category</th></tr></thead><tbody><tr><td>Aarav</td><td>65000.00</td><td>Medium Salary</td></tr><tr><td>Vivaan</td><td>72000.00</td><td>Medium Salary</td></tr><tr><td>Aditya</td><td>58000.00</td><td>Low Salary</td></tr><tr><td>Arjun</td><td>81000.00</td><td>High Salary</td></tr><tr><td>Kabir</td><td>69000.00</td><td>Medium Salary</td></tr><tr><td>Reyansh</td><td>62000.00</td><td>Medium Salary</td></tr><tr><td>Ananya</td><td>52000.00</td><td>Low Salary</td></tr><tr><td>Diya</td><td>48000.00</td><td>Low Salary</td></tr><tr><td>Myra</td><td>61000.00</td><td>Medium Salary</td></tr><tr><td>Sara</td><td>55000.00</td><td>Low Salary</td></tr><tr><td>Ishita</td><td>47000.00</td><td>Low Salary</td></tr><tr><td>Meera</td><td>59000.00</td><td>Low Salary</td></tr><tr><td>Rohan</td><td>75000.00</td><td>High Salary</td></tr><tr><td>Karan</td><td>68000.00</td><td>Medium Salary</td></tr><tr><td>Nikhil</td><td>83000.00</td><td>High Salary</td></tr></tbody></table>			emp_name	salary	salary_category	Aarav	65000.00	Medium Salary	Vivaan	72000.00	Medium Salary	Aditya	58000.00	Low Salary	Arjun	81000.00	High Salary	Kabir	69000.00	Medium Salary	Reyansh	62000.00	Medium Salary	Ananya	52000.00	Low Salary	Diya	48000.00	Low Salary	Myra	61000.00	Medium Salary	Sara	55000.00	Low Salary	Ishita	47000.00	Low Salary	Meera	59000.00	Low Salary	Rohan	75000.00	High Salary	Karan	68000.00	Medium Salary	Nikhil	83000.00	High Salary
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...

7. ORDER BY

Display employees in descending order of salary.

```
SELECT emp_name, salary
FROM Employee
ORDER BY salary DESC;
```

OUTPUT

I/OAI Agent86 ms

STDIN

Input for the program (Optional)

Output:

emp_name	salary
Nikhil	83000.00
Arjun	81000.00
Rudra	76000.00
Rohan	75000.00
Dhruv	73000.00
Vivaan	72000.00
Manav	71000.00
Navya	70000.00
Kabir	69000.00
Karan	68000.00
Atharv	67000.00
Riya	66000.00
Aarav	65000.00
Veer	64000.00
Kiara	63000.00

8. JOIN Employee, Department and Project

Display employee name, department, project and salary.

```
SELECT
    e.emp_name,
    d.dept_name,
    p.project_name,
    e.salary
FROM Employee e
JOIN Department d
ON e.dept_id = d.dept_id
JOIN Project p
ON e.project_id = p.project_id
ORDER BY d.dept_name, e.emp_name;
```

OUTPUT

 I/O  AI Agent 83 ms 

STDIN

Input for the program (Optional)

Output:

emp_name	dept_name	project_name	salary
Dev	Finance	Financial Dashboard	56000.00
Karan	Finance	Financial Dashboard	68000.00
Manav	Finance	Financial Dashboard	71000.00
Nikhil	Finance	Financial Dashboard	83000.00
Rohan	Finance	Financial Dashboard	75000.00
Yash	Finance	Financial Dashboard	62000.00
Ananya	HR	Recruitment Portal	52000.00
Diya	HR	Employee Wellness	48000.00
Ishita	HR	Recruitment Portal	47000.00
Meera	HR	Employee Wellness	59000.00
Myra	HR	Recruitment Portal	61000.00
Sara	HR	Employee Wellness	55000.00
Aarav	IT	Cloud Migration	65000.00
Aditya	IT	Mobile Application	58000.00
Ariun	IT	Cloud Migration	81000.00

RESULT

The Employee–Department–Project database was successfully created with:

- 30 Employees
- 5 Departments
- 8 Projects

The following SQL concepts were successfully demonstrated:

1. Selection
2. Projection
3. Aggregate Functions
4. GROUP BY
5. HAVING
6. CASE Expression
7. ORDER BY
8. JOIN